

Creating Conversational Datasets for Retrieval-Augmented Generation Applications is Hard: Challenges & Research Opportunities

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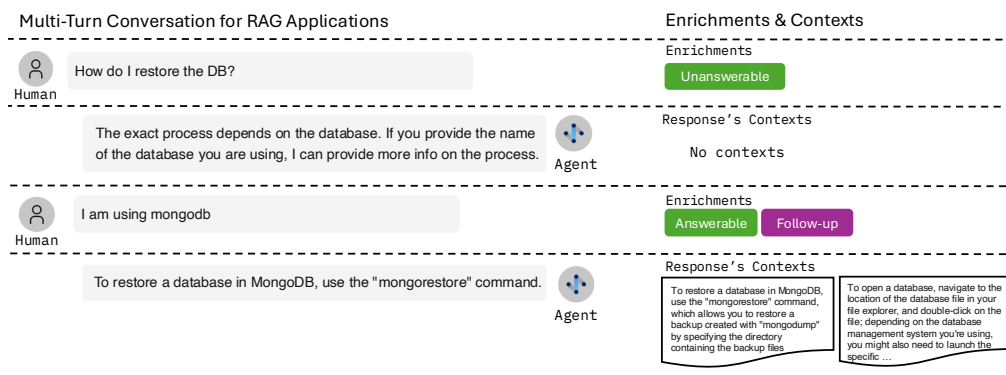


Figure 1: Example of a multi-turn conversation for a RAG application, where each turn may be enriched with different types and each agent turn is associated with relevant contexts.

Abstract

Retrieval-augmented generation (RAG) has been proven to help mitigate hallucinations from large language models (LLMs). However, as more domains adopt this method, the need for human-created conversational data increases, as human-created conversations are naturally driven better. Yet, in our experience, when we tasked several annotators to create such data for RAG applications, we learned that creating conversational data for RAG is hard, leading to cases where we had to reject 35% of data in some batches. In this paper, we interview a group of annotators to understand what makes this task challenging. We distill insights from the formative study and outline potential future directions in the intersection of RAG applications and HCI.

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CCS Concepts

• **Human-centered computing** → **Interactive systems and tools**; • **Applied computing** → *Business intelligence*; *Enterprise applications*.

Keywords

Retrieval-Augmented Generation, Data Creation, Data Annotation

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1 Introduction

Large language models (LLMs) like GPT4¹ and Llama 3² recently became popular but still face issues, especially hallucination [14]. Retrieval-augmented generation (RAG) [21, 34] has been proven a popular approach for mitigating these issues as it uses (1) a retriever

¹<https://openai.com/index/gpt-4/>

²<https://www.llama.com/>

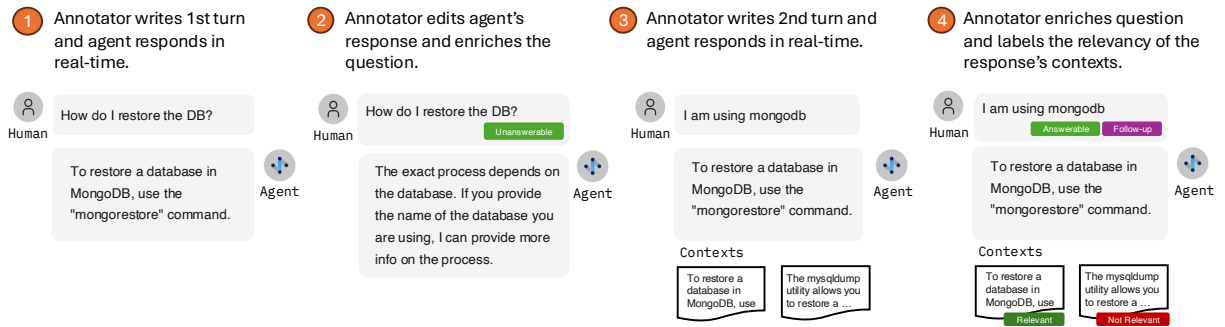


Figure 2: Example flow of how an annotator would create a RAG conversation.

model to retrieve contexts relevant to the particular user question and (2) a generator model that then generates a response grounded in the retrieved contexts. This two-stage process is popular with enterprises that typically need to interface with domain datasets in a conversational manner, usually in the form of a chatbot. Thus, conversational datasets, such as DialogStudio [39], RAGBench [9], and MTRAG [17], are important for training and testing RAG applications. As more domains adopt RAG solutions, the need for conversational RAG data increases, especially those created by humans [1]. However, asking humans to create conversations, specifically for RAG, is challenging.

What distinguishes a human-to-human conversation from a human-to-agent conversation (specifically an agent with a RAG setup) is primarily in the retrieved contexts attached to each of the agent’s responses (see an example in Figure 1). This structure introduces several requirements (1) at the global level: the conversation must flow coherently and naturally with good style and (2) at the retrieval level: each of the retrieved contexts needs to be relevant to the attached agent’s response and (3) at the generator level: the agent’s response is faithful and grounded to the contexts.

Our initial observations on the challenges of creating conversational RAG datasets began when we manually sifted through the data from our internal annotators and annotators from an unnamed external provider. We tasked them to create multi-turn RAG conversations (each one 6 to 8 turns) across several domain corpora. Annotators on average took several hours³ to create a single conversation as the task involved reading at least 3 passages per turn and usually many more. In one batch containing approximately 200 conversations that we manually reviewed, we accepted only 65% of the conversations.

We conducted a formative study to understand the aspects of creating RAG data our non-technical annotators struggled with. We conducted the study on an internal tool, RAGDATA CREATOR, which enables users to create conversational RAG datasets with a real-time component that visualizes the outputs of the retriever and generator models immediately after a human annotator interacts with it, subsequently, allowing the annotator to edit their outputs. Our study finds that creating *high-quality* conversations for RAG is challenging because

- (1) It is a complex task involving several inter-connected sub-tasks that cannot be broken down into smaller tasks.
- (2) It is a task that annotators mentally perceive as different from how humans naturally converse, which can potentially derail them from making the conversation flow coherently and naturally.
- (3) It involves additional effort on the human annotators to tinker with the individual retriever and generator models and make edits to undesired outputs (i.e. fixing the agent’s response and its attached contexts).

Figure 2 showcases the workflow of an annotator creating a RAG conversation. An annotator needs to be cognitive of the changes done at a per-turn level—such as fixing the agent’s response to ask a clarification question (Step 2)—can impact the subsequent turns of how the conversation would unfold. At a per-turn level, an annotator also needs to manage a multitude of information all the while fixing the issues from the outputs of the generator and retriever models by reading the contexts, adding more relevant contexts if needed, and fixing the generator’s response to include all the relevant contexts (Step 4).

Our work makes the following contributions:

- A formative study conducted on 7 experienced annotators to help us understand why creating multi-turn conversations for RAG is challenging.
- Specific challenges human annotators face when creating conversational RAG datasets.
- Insights and potential research directions in the intersection of HCI and RAG applications.

2 Related Works

Existing works in the intersection of RAG applications and HCI are focused on the specific use case it intends to support, such as WaterBot, a chatbot with a RAG architecture that helps residents of Arizona understand water resources [29]. In another related work, Zhao et al. [40] introduces a categorization of the different types of RAG tasks based on the external data and goals of the task. They discuss challenges introduced at each level of task difficulty, which may help us understand why some conversations might be more difficult for the RAG application to handle. However, it does not tackle the challenges that arise when humans create RAG conversations. More broadly, existing works on RAG applications focus on

³We do not report specific numbers due to company policy.

(1) the systems aspect of the RAG application or (2) the datasets for RAG applications, including tooling that enables humans to create the data and methods that synthetically generate data.

2.1 Systems aspect of RAG applications

The systems aspect concerns the individual modules and architectural aspects of the RAG application (i.e. retrieval and generator models) [5, 8, 15, 26]. Selfmem [5] is an example of an architectural improvement that boosts RAG performance by using self-memory that selects data closest to the current inputs. DIRAS [26] is an example of improving a particular part of RAG—specifically, context relevancy—by fine-tuning a model that annotates the relevancy of the retrieved contexts. MeMemo [38] focuses on the production aspects of RAG, specifically a toolkit that enables developers to have a RAG setup in-browser rather than the traditional complex backend setup. While improving the systems aspect of RAG is important, creating and obtaining the data in the first place is also key.

2.2 RAG Datasets

Many RAG systems are evaluated against existing RAG benchmarks, such as RAGAs [7], RAD-Bench [20], RAGBench [9], RGB [4], and MTRAG [17]. Some datasets are synthesized and augmented, which is especially useful for low-resource domains [22, 27, 33, 35, 36]. DialogStudio [39] offers a diverse collection of dialogue datasets, including external data attached to a conversation, but their experiments on the datasets focus only on the generative aspect and not the retrieval aspect. Datasets generated by Converser [13] occur at a per-turn level, where the synthesized questions do not necessarily depend on each other the same way naturally human-generated questions unfold in conversations—it would not be able to generate a question that asks the agent for more information based on the context in the previous turns (similar to the agent’s turn in Figure 1). Notably, MTRAG [17] contains 110 multi-turn human-created conversations used for testing the full RAG pipeline. This dataset helps RAG pipelines understand their performance on longer multi-turn conversations (averaging 7 turns), unanswerable questions, and non-standalone questions that depend on the previous turns. We envision similar, human-created datasets are needed as we see more customized RAG applications on niche datasets. This requires us as a research community to understand what makes creating such datasets challenging.

2.3 Human-Created Data and Annotation

Another major related thread to our work includes past work on general data creation and data annotation [10, 11, 19, 24, 30]. Many commercial tools exist for creating and curating data, including Mechanical Turk (MTurk)⁴, Appen⁵, Prodigy⁶, etc. Some of them, such as BasicAI⁷, Label Studio⁸, and Labelbox⁹, also offer the ability to create and/or annotate different aspects of conversational data (e.g., rate dialogues/responses, add metadata, tag entities present in the

dialogue, etc.). However, none of these tools enables annotators to create conversational multi-turn RAG datasets with real-time agent response generation and context retrieval that RAGDATACREATOR enables.

Tooling is important for minimizing errors in the dataset by human annotators [28, 37], especially with the complexity of the underlying RAG dataset structure. Joko et al.’s work [16] explored different UIs for annotating entities in conversations, specifically for conversational entity linking. These UIs were experimented on MTurk with a Wizard of Oz setup, which does not have a real-time aspect that would have enabled developers to collect on-the-spot feedback data regarding outputs of the retriever and generator model. Panagiotis et al.’s work explores how conversational interfaces can be effective for crowdsourced micro-tasks such as annotating images [24]. However, it is still unclear how conversational interfaces help in complex tasks such as creating conversational RAG data.

3 Creating Conversational Datasets for RAG Applications

This section introduces the typical flow of how an annotator would create a multi-turn conversation for RAG. A client engineer is tasked with creating a multi-turn RAG dataset for a niche domain. The client engineer instructs a team of annotators to create the dataset using a tool like RAGDATACREATOR. An annotator would begin creating a conversation by inputting the settings for the retriever and generator, specified by the client engineer’s instructions. These settings include prompt format and model type. RAGDATACREATOR can communicate with any model available through an API.

In RAGDATACREATOR, creating a conversation occurs at a per-turn level, similar to how human-to-human conversations naturally occur. At a per-turn level, an annotator, Alice, does the following:

- (1) **Write a question to the agent** (using the chat bar at the bottom of Figure 3). The agent automatically responds using the selected generator and retriever models. As requested by the client engineer’s instructions, Alice enriches the question with labels, such as indicating whether the question is standalone or not, or whether it is a clarification question. The enrichment types are customizable according to the desired dataset.
- (2) **Read and inspect the contexts** that were attached to the agent’s response (see Figure 3). By default, each agent response is accompanied by three contexts from the retriever. Alice also marks the relevancy of each context to the agent response. Sometimes, due to a fault in the retriever model, the retrieved contexts are irrelevant to the agent response. Alice needs to add more relevant contexts by using the **RE-QUERY TOOL** (right-hand side of Figure 3). The **RE-QUERY TOOL** helps her find contexts based on the keywords she enters into the search bar. Alice reviews and adds more relevant contexts to the response of the current turn.
- (3) **Edit the agent’s response** to include core aspects from the contexts. Since Alice updated the contexts of the agent response, the agent response must be regenerated so that it is current to the newly attached contexts. Alice clicks on the regenerate button (see Figure 3) and the generator overwrites

⁴<https://www.mturk.com/>.

⁵<https://www.appen.com/>.

⁶<https://prodigy.ai/>

⁷<https://www.basic.ai/basicai-cloud-data-annotation-platform/large-language-model-and-generative-ai-data-annotation-toolset>

⁸https://labelstud.io/templates/dialogue_analysis

⁹<https://labelbox.com/solutions/conversational-ai/>

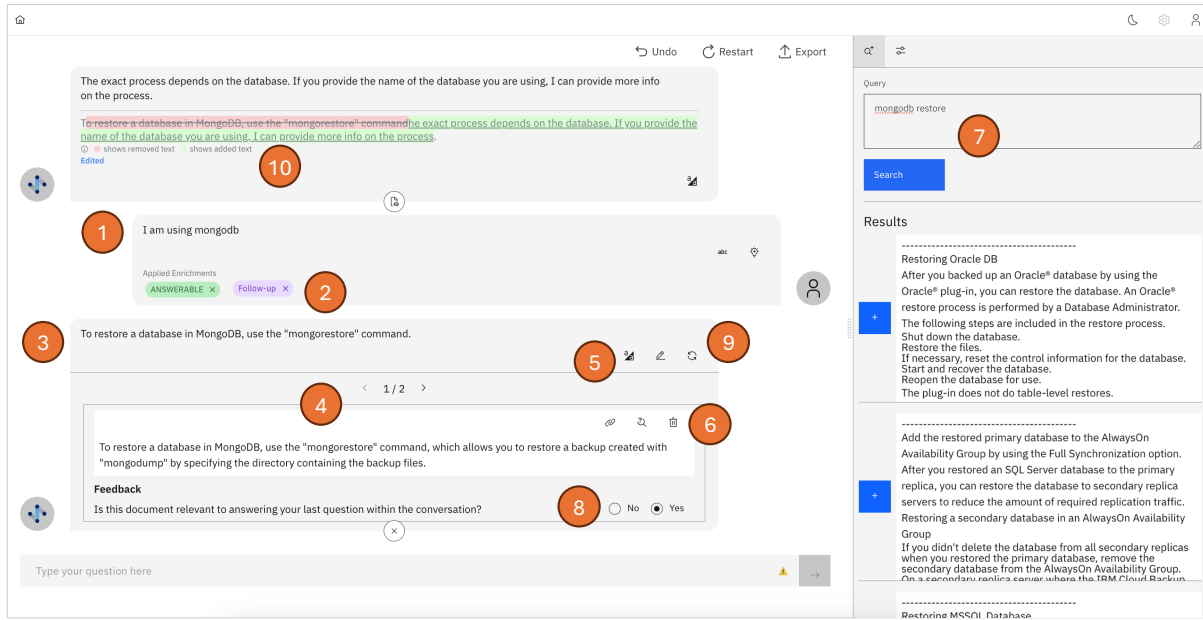


Figure 3: RAGDATACREATOR UI for Creating Conversational Datasets for Retrieval-Augmented Generation (RAG) Applications: The left-hand side contains a view of the conversation. The turn in (1) is the annotator’s query to the agent. The annotator enriches the query (2). The next turn is the agent’s turn (3), along with the contexts attached in-situ (4). The annotator navigates through each of the contexts using the arrow buttons. The annotator can view what parts of the document overlap with the contexts using the context overlap button in (5). The annotator can also delete irrelevant contexts using the button in (6) and also add more relevant contexts by searching for similar documents in the **REQUEST TOOL** in (7). Every attached context to a response must be provided with feedback using the form in (8). The attached contexts in the created conversations are essential to help improve the retrieval aspect of the RAG system. Moreover, any changes to the set of contexts must be accompanied by the annotator regenerating the agent’s response by clicking on the regenerate button in (9). Next to the regenerate button is the edit button that edits the agent’s response. The annotator must also edit the agent’s response to help improve the generation part of the RAG system. A visualization of the changes to the original agent’s message is visualized underneath the edited response (10).

its old response with an updated response based on the current state of the contexts. The agent’s new response now includes snippets from the contexts Alice added. Sometimes, due to a fault in the selected generator model, the agent response does not include the core ideas in some of the retrieved contexts. Alice must edit the generated response by clicking the edit button (see Figure 3) so that the response also integrates some parts of the ignored contexts.

Once Alice completes editing the current user and agent turn, she creates another question based on the previous set of turns to make a multi-turn, cohesive conversation. Once several turns of the conversation (based on the instructions of the client engineer) are made, Alice reviews her conversation using the left-hand side of the screen in Figure 3. Finally, she exports her conversation using the “Export” button at the top panel.

By design, previous turns in a conversation cannot be edited, as it would disrupt how conversations naturally occur i.e. they progress over time without edits to past turns. Moreover, if a question in a past turn is edited, the questions in the next turn may not make sense. However, users can *undo* an entire turn which erases the

previous turn (including the agent’s response, contexts, and the user’s question).

4 Qualitative Evaluation

We wanted to understand what made creating data for RAG applications challenging for annotators. We ran a formative study involving semi-structured interviews with 7 internal annotators (denoted as U1-U7). Our internal annotators (1 man, 6 women) are considered power users of RAGDATACREATOR, having been using it for 6 months straight at the time of conducting the study. All annotators, specifically highly skilled in annotation, are hired solely to perform language annotation tasks and are paid well above minimum wage.

During the study, we asked annotators to walk us through the steps on how they would use RAGDATACREATOR to create a multi-turn conversation. We asked them to speak their thoughts aloud to understand their thought process. In total, the study took approximately half an hour for each participant. To avoid bias among the participants, we scheduled individual sessions with each participant and we screen-recorded the interviews with consent from the

participants. We anonymized the transcripts and performed inductive thematic analysis [3] on the data to identify emerging patterns across participants, resulting in 3 overarching themes, which we summarize as findings in the following sections.

4.1 Sub-tasks are interconnected to each other and increase task complexity

The task of creating a conversation is complex, as U2 says, “There are several layers to the task... By the time I’ve changed enrichments and changed relevance [of the contexts to a particular response] and edited the answer, you know, I’m basically losing track of what I did.” Moreover, the task cannot be broken down into smaller pieces, requiring annotators to juggle between many different aspects of the data, such as ensuring context diversity, the faithfulness of the agent’s responses to the contexts, the completeness and appropriateness of the agent’s responses to the question, and the coherency of the whole conversation’s flow.

The amount of reading annotators are instructed to do as part of instructions adds to the task’s complexity. Analyzing the interview screen recordings, we observed that all the annotators (U1-U7) began reading all the response’s contexts—each one often more than 100 words long—as the first immediate step after receiving a response to the question they created. And at almost every turn, especially when the client engineer requires annotators to use a primitive (an early-stage) version of the retriever model, annotators have to use the `REQUERY TOOL` to find new contexts, and they have to read the contexts returned by the `REQUERY TOOL`. U4 says, “Well, the first thing I do [is] read the passages... We need to read all this to see whether information is good or not to add to my [agent’s] answers.” U6 also echoes similar statements, “So now I’m reading all the questions and all the passages to see which one can go.”

The amount of reading makes some annotators feel like they are “in constant cognitive overload” (U2, U5). U2 continues, “There just comes a point where my brain melts down from reading through so much material and you’re trying to separate the wheat from the chaff.” Interestingly enough, the same annotator also likes the UI with all of the text on one screen (“I like being able to see as much of it [contexts, response, question] as possible.”). This shows how inseparable the various parts of the conversation are to each other and they cannot be broken down into smaller tasks. U3 even says, “You have to read all the the information in the documents. And, sometimes that already gives you an idea for the next question, if you find some interesting text.” These statements illustrate the cognitive overload in reading and tracking a multitude of information when annotators create conversational data for RAG.

4.2 Building conversations for RAG applications is different from how humans naturally converse

There is a mismatch in the mental model between how people naturally speak conversations and how a RAG application would need to participate in a conversation. When creating conversations in `RAGDATACREATOR`, edits to previous turns cannot be done. In general, people who are chatting with chatbots usually do not request editing a past turn. If there are troubles in a conversation, humans usually resolve them in the future or current turn

by clarifying any misunderstanding the agent had regarding their intentions [25, 31, 32].

In `RAGDATACREATOR`, there is a real-time aspect to creating conversations, where agents respond to the annotator’s questions on the spot. However, annotators are still compelled to edit questions in past turns, sometimes due to typos (U3, U4, U6). U4 described instances where she wanted to edit her questions to the agents from a past turn, “I review [the entire conversation] before I export. I [see] a mistake: This question doesn’t look so good. I would like to change the question. [...] I want to go back and correct my mistakes and [not] lose the conversation, because there’s a lot of time and effort doing this.” Moreover, typos in conversational datasets may even be desirable as they help the trained model be more robust to these noise users usually make.

While users of general chatbots do not request to edit past turns, annotators in `RAGDATACREATOR` do not perceive creating conversations for RAG the same as naturally speaking a conversation as a human would, even though that is the goal. Creating conversational RAG datasets with attached contexts is perceived as creating complex, written artifacts such as essays, where people typically want to edit past texts and past actions at a future time.

4.3 Generator and retriever model performance introduces burdens to annotator at a per-turn level

In the earlier stages of model development, client engineers request annotators to use these early-stage models to help create additional training data and discover issues with the retriever and generator models. Weaknesses in the retriever or generator model introduces burdens on the annotator to provide data to help improve the respective models.

In the past, the annotators in this study have been tasked to use an early-stage version of the retriever, and during this study, they voiced their concerns regarding the added burden (U1, U2, U3, U5, U7). The early-stage retriever had an issue where the set of documents returned by a query was perceived as not “diverse”, i.e. the returned documents had similar texts and did not differ in content much. U7 says, “The tool [(`RAGDATACREATOR`)] works well but the contexts are very very limited and mostly the same. It is very unpleasant to work with.”

Because of the retriever model, annotators search for more contexts using the `REQUERY TOOL`, forcing them to interact with the retriever model more. U2 says, “Coming up with requerying keywords is challenging. The results are VERY busy, maybe having 9 return hits, including duplicates, is too many all at once.” U7 connects this with the context’s diversity, “It’s very difficult to come up with the right [set of contexts]... You ask us for diversity of the passages [but the retriever model] makes it harder.”

In some cases, the annotators encounter turns for which early-stage retrievers do not return any relevant contexts. U3 says, “I got to a dead end. I had research like 20 different [keywords]. And I never got a [relevant context], so I had to completely change my my idea about how to to handle that conversation... Yeah, it’s just, you know, it loses creativity in the conversation. That’s the pity.” U3 also illustrates a concrete example, “The seed question was something about bone marrow. And I had already figured out in my head that

I could go from bone marrow to which doctor specialty. Could they treat this? Where could he go to school? And I went on Google and looked at Wikipedias and found the specialty and what was needed, and the colleges they had to go to... which medical schools? In which hospitals? [...] But when I went into the tool and tried to find those documents, there [were none].” While from a model development perspective, the developers managed to gather more test cases of model failure to fix, annotators are burdened with the added sub-task of essentially testing the model on top of creating a conversational dataset for the RAG application.

Early-stage generators also burden the annotators. Lower-quality generated responses such as incomplete answers force the annotators to rewrite the response. Some annotators have voiced that the generated responses from the early-stage generator do not seem to change even when the state of the contexts change. U1 says, “If I don’t like the [agent’s] answer, I regenerate [by clicking on the regenerate button]. But it just keeps showing the same answer over and over. [...] I thought [that] if I keep clicking regenerate, AI will generate [a] new response for me, like [a] different response.” While annotators almost always edit the generator’s response, the annotators in our interviews did not voice as many concerns regarding early-stage generators as much as the issues stemming from early-stage retrievers.

In short, there are complexities at a per-turn level introduced by the weaknesses in the retriever and generator models. In early-stage retriever models, the limited passages returned can limit the annotator’s creativity and waste their time reading duplicate documents. The model’s performance impacts the annotator’s task load.

5 Discussion

Given our findings, we now distill insights and potential research directions for the HCI research community.

5.1 Need for design guidelines and principles for RAG data creation tools

Past research has shown that the UI can impact the annotator’s performances [28, 37]. However, there is little to no research on *how* to best design interfaces for creating RAG data. We need to build better tooling to support the workflow for creating data for RAG. Standard designs of chat interfaces work great for conversing naturally, but, as shown in our interviews, they may conflict with the human annotator’s mental model of *building* a RAG conversation. There are opportunities for HCI research to hone in on the best design practices when it comes to tooling for RAG data creation, similar to past HCI research on the best practices for designing toolkits [23], dashboards [2], UIs for annotating conversations [16], chatbots [6], and conversational interfaces for crowdsourced microtasks [24].

5.2 Need best practices for tasking annotators to create RAG data

A whole breadth of work has been dedicated to the best practices for formulating tasks on crowdsourcing platforms [18], which include breaking up complex work into smaller tasks [19]. However, conversations cannot be broken into MapReduce-like operations

as conversational structures are inherently different. Parts of conversations are not created in parallel since conversation occurs in sequences and progresses through time [25]. Past works have also explored the best practices for formulating tasks and guidelines for annotators across different tasks, such as evaluating visualizations [12] and gathering design feedback for user interfaces [10]. We should draw on these works and understand how we can leverage them for tasking human annotators for RAG data creation.

6 Conclusion

Through an initial formative study, we explore the challenges human annotators face when tasked to create conversational RAG data. We discover that (1) the task is complex involving several interconnected sub-tasks, (2) human annotators perceive the task as mentally different than how humans converse naturally with other humans, and (3) human annotators have the added burden of fixing issues in the outputs from the individual generator and retriever model. As more domains adapt RAG applications for their use cases, there is a growing need for the best practices regarding tooling design and formulating annotator tasks specifically for creating RAG data.

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